

# PHASE 7 – PROJECT DOCUMENTATION

## Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

### 7.1 Introduction

Project Documentation is the process of recording all important information about the project, including the technologies used, implementation details, testing results, and project outcomes. Proper documentation helps future users understand how the ServiceNow application was developed and how the ACLs were configured.

### 7.2 Technologies Used

| Technology                  | Purpose                              |
|-----------------------------|--------------------------------------|
| ServiceNow                  | Platform for application development |
| JavaScript                  | Script-Controlled Read ACL           |
| Access Control Lists (ACLs) | Record-level security                |
| Role-Based Access Control   | Permission management                |
| GitHub                      | Project version control              |
| Google Drive                | Demo video sharing                   |

## 7.3 Project Features

The application includes the following features:

- Custom Institution Details table.
- User creation and role assignment.
- Four custom roles (bb1, bb2, bb3, bb4).
- Read ACL with JavaScript.
- Create ACL.
- Write ACL.
- Delete ACL.
- Impersonation-based testing.

## 7.4 Security Implementation

The project implements role-based security by combining user roles with record conditions.

| ACL    | Purpose               |
|--------|-----------------------|
| Read   | View only EEE records |
| Create | Allow new records     |
| Write  | Edit records          |
| Delete | Delete records        |
| Admin  | Full access           |

## **7.5 Advantages**

- Protects confidential student records.
- Prevents unauthorized access.
- Demonstrates practical ServiceNow security.
- Supports secure record management.
- Easy to manage using custom roles.

## **7.6 Limitations**

- The project is limited to the EEE branch condition.
- Requires proper role assignment.
- Depends on ServiceNow permissions.
- Additional conditions may require extra scripting.

## **7.7 Future Enhancements**

The application can be improved by adding:

- Department-wise dynamic permissions.
- Automatic role assignment.
- Email notifications.
- Audit logging.
- Approval workflows.

## **7.8 Documentation Screenshots**

Add 2–3 screenshots here:

1. Institution Details Table

dev395038.service-now.com/now/nav/ui/classic/params/target/sys\_db\_object\_list.do%3Fsysparm\_userpref\_module%3D7e7ca89ac0a8000901594ba32405461%26sysparm\_q...

serviceNow

Tables

Search

Table Institution Details

\* Label

Institution Details

\* Name

u\_institution\_details

Extends table

Application

Global

Create module

☒

Create mobile module

☒

Add module to menu

-- Create new --

New menu name

Remote Table

☐

Columns

Controls

Application Access

Table Columns

for text

Search

New

Dictionary Entries

| Column label | Type          | Reference | Max length | Default value | Display |
|--------------|---------------|-----------|------------|---------------|---------|
| Updated by   | String        | (empty)   |            | 40            | false   |
| Updates      | Integer       | (empty)   |            | 40            | false   |
| Sys ID       | Sys ID (GUID) | (empty)   |            | 32            | false   |

Table - Institution Details

Search

Delete

Update

Delete All Records

Table Institution Details

Dictionary Entries

| Column label        | Type          | Reference | Max length | Default value | Display |
|---------------------|---------------|-----------|------------|---------------|---------|
| Updated by          | String        | (empty)   |            | 40            | false   |
| Updates             | Integer       | (empty)   |            | 40            | false   |
| Sys ID              | Sys ID (GUID) | (empty)   |            | 32            | false   |
| Student Roll Number | String        | (empty)   |            | 40            | false   |
| Faculty Name        | Reference     | User      |            | 32            | false   |
| Updated             | Date/Time     | (empty)   |            | 40            | false   |
| Student Name        | Reference     | User      |            | 32            | false   |
| Branch              | Choice        | (empty)   |            | 40            | false   |
| Created by          | String        | (empty)   |            | 40            | false   |
| Email               | String        | (empty)   |            | 40            | false   |
| Description         | String        | (empty)   |            | 40            | false   |
| Phone Number        | String        | (empty)   |            | 40            | false   |
| Created             | Date/Time     | (empty)   |            | 40            | false   |

Insert a new row...

Type here to search

35°C Sunny

14:43

27-09-2026

dev395038.service-now.com/now/nav/ui/classic/params/target/sys\_db\_object.do%3Fsys\_id%3D2ae8930c3630f10949353f1b4013106%26sysparm\_record\_target%3Dsys\_db\_...

servicenow All Favorites History Admin Table - Institution Details Search Delete Update Delete All Records

Table Institution Details Explore REST API

Access Controls (8) Security Data Filters Labels (1) Database Indexes (3) Table Subscription Configuration (1)

Updated Search Actions on selected rows...

|  | Name                  | Decision Type | Operation | Type   | Active | Updated by | Updated             |
|--|-----------------------|---------------|-----------|--------|--------|------------|---------------------|
|  | u_institution_details | Allow If      | delete    | record | true   | admin      | 2026-09-22 20:04:05 |
|  | u_institution_details | Allow If      | write     | record | true   | admin      | 2026-09-22 20:02:40 |
|  | u_institution_details | Allow If      | create    | record | true   | admin      | 2026-09-22 20:00:32 |
|  | u_institution_details | Allow If      | read      | record | true   | admin      | 2026-09-22 19:58:04 |
|  | u_institution_details | Allow If      | delete    | record | true   | admin      | 2026-09-22 19:37:29 |
|  | u_institution_details | Allow If      | write     | record | true   | admin      | 2026-09-22 19:37:29 |
|  | u_institution_details | Allow If      | create    | record | true   | admin      | 2026-09-22 19:37:29 |
|  | u_institution_details | Allow If      | read      | record | true   | admin      | 2026-09-22 19:37:29 |

1 to 8 of 8

## 2. Read ACL Configuration

dev395038.service-now.com/now/nav/ui/classic/params/target/sys\_security\_acl.do%3Fsys\_id%3D24e31d38c3630f10949353f1b4013104%26sysparm\_record\_target%3Dsys\_se...

servicenow All Favorites History :Access Control - u\_institution\_details Search

Access Control u\_institution\_details

Type record Application Global

Operation read Active ☒

Admin overrides ☒ Advanced ☒

Decision Type Allow If

Protection policy -- None --

Name u\_institution\_details

Description Allow read for records in u\_institution\_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u\_branch=EEE\*EQ) evaluates to true.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

1 to 1 of 1

### 3. Final Working Screen (EEE User or Admin)

The screenshot displays the ServiceNow 'Institution Details' interface. The browser's address bar shows the URL: `dev395038.servicenow.com/now/nav/ui/classic/params/target/instance_details_list.do%3Fsysparm_clear_stack%3Dtrue`. The ServiceNow header includes the logo and navigation links: All, Favorites, History, and a search bar. Below the header, there's a filter bar with 'Institution Details' selected and a search input. The main content area shows a table with the following data:

| Branch | Description     | Email | Faculty Name      | Phone Number | Student Name    | Student Roll Number |
|--------|-----------------|-------|-------------------|--------------|-----------------|---------------------|
| EEE    | ACL Is Executed |       | Alejandra Perault |              | Alyson Gilligie | STU200              |

The bottom of the screen shows a Windows taskbar with the search bar, task icons, system tray (32°C Sunny, 10:28, 27-09-2024), and a pagination bar indicating '1 to 1 of 1'.